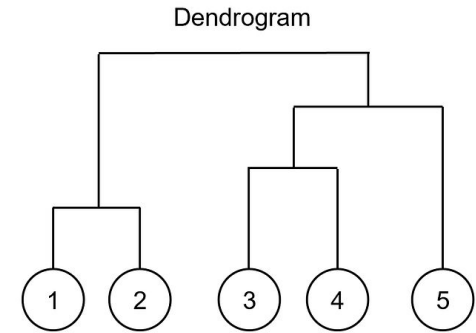
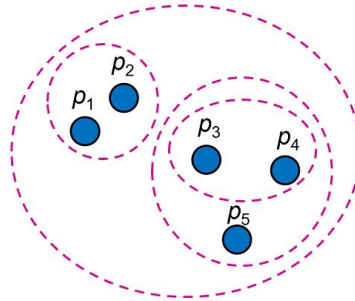
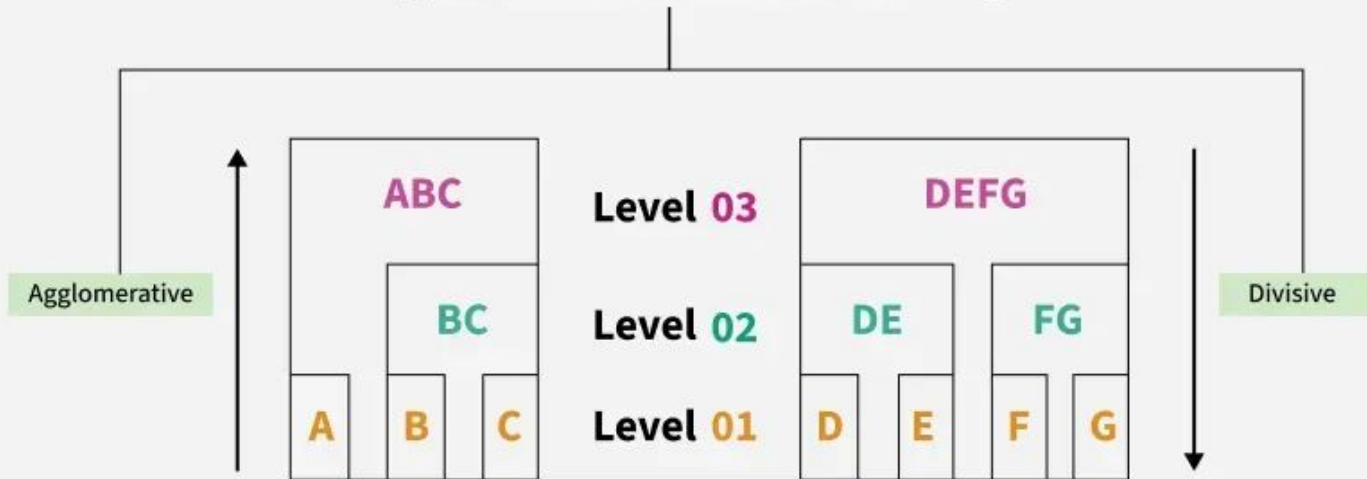


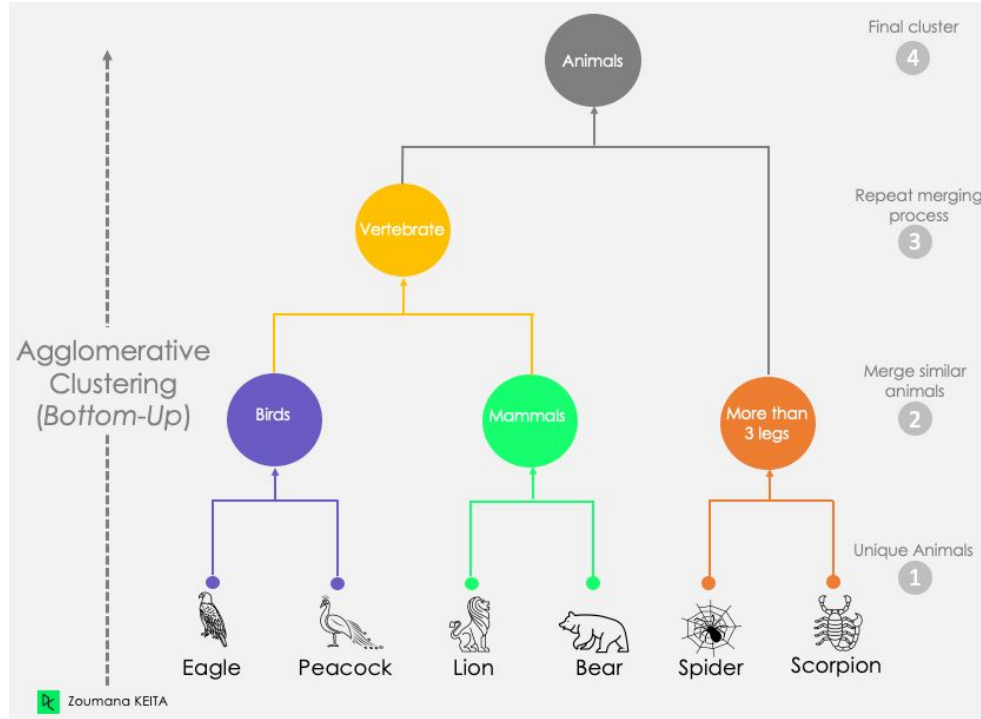
Hierarchical Clustering and DBSCAN

Hierarchical clustering is an unsupervised machine learning method that groups data points into a tree-like hierarchy of clusters based on their similarity

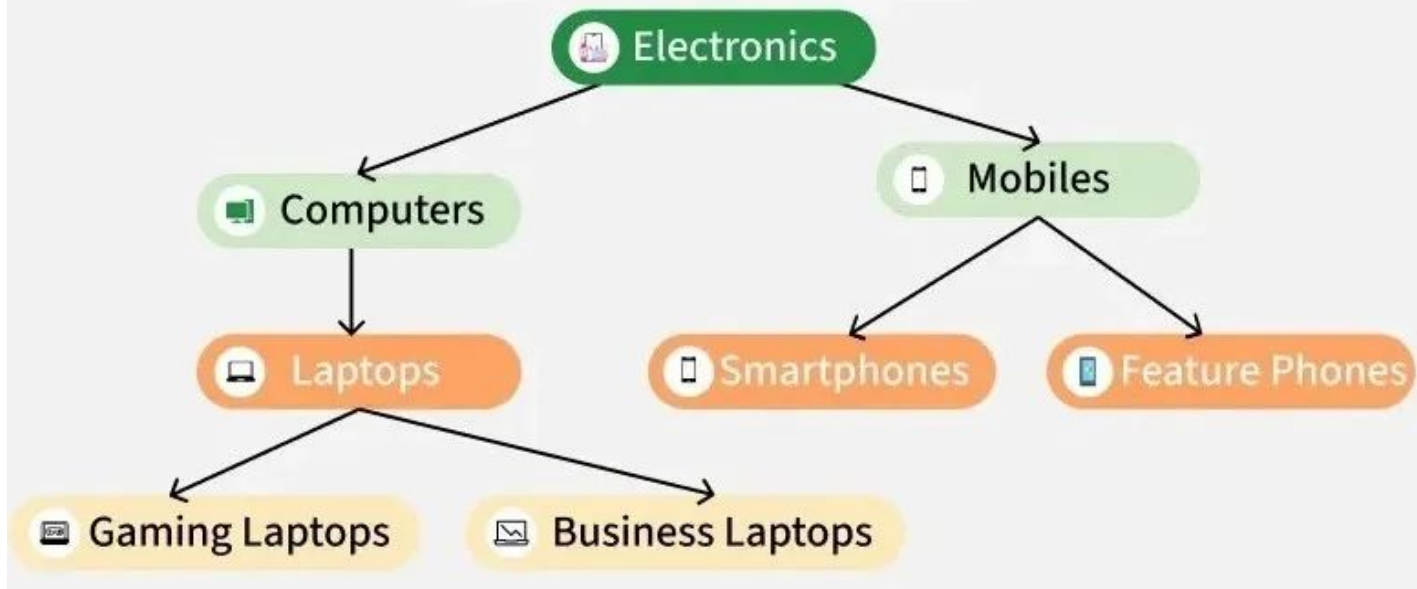


Types of Hierarchical Clustering



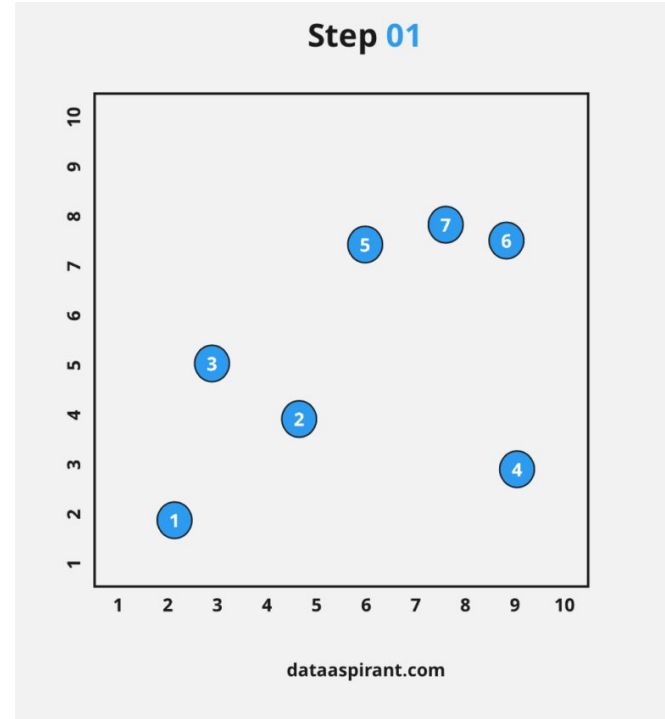


Product Categorization Tree



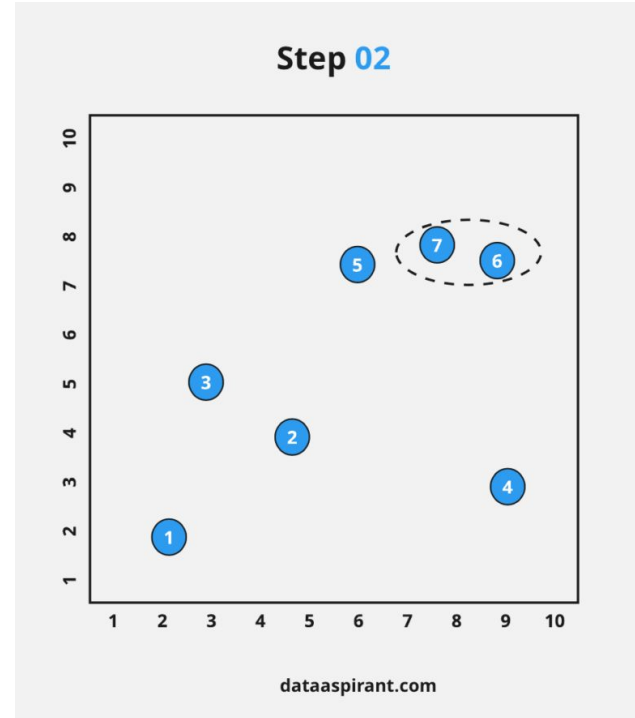
Step 1

First, make each data point a “single - cluster,” which forms N clusters. (let’s assume there are N numbers of clusters).



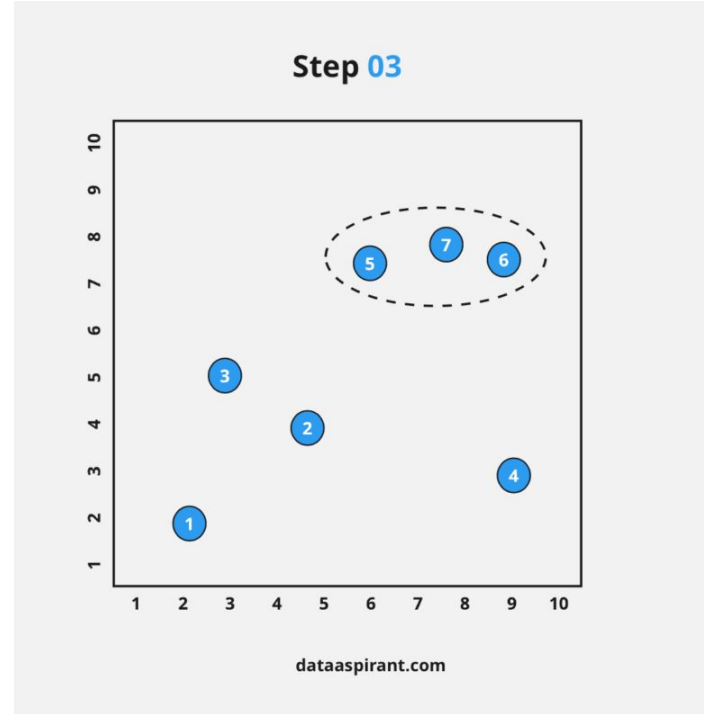
Step 2

Take the next two closest data points and make them one cluster; now, it forms **N-1** clusters.



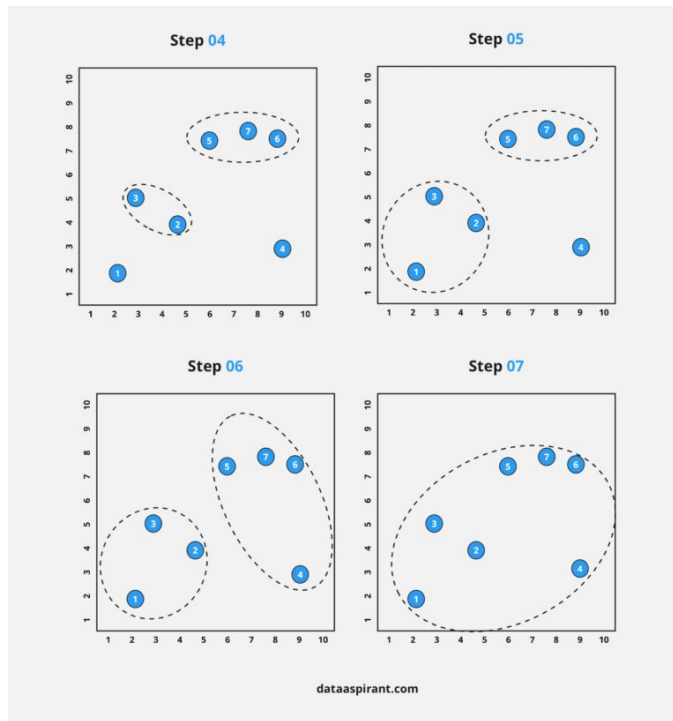
Step 3

Again, take the two clusters and make them one cluster; now, it forms N-2 clusters.

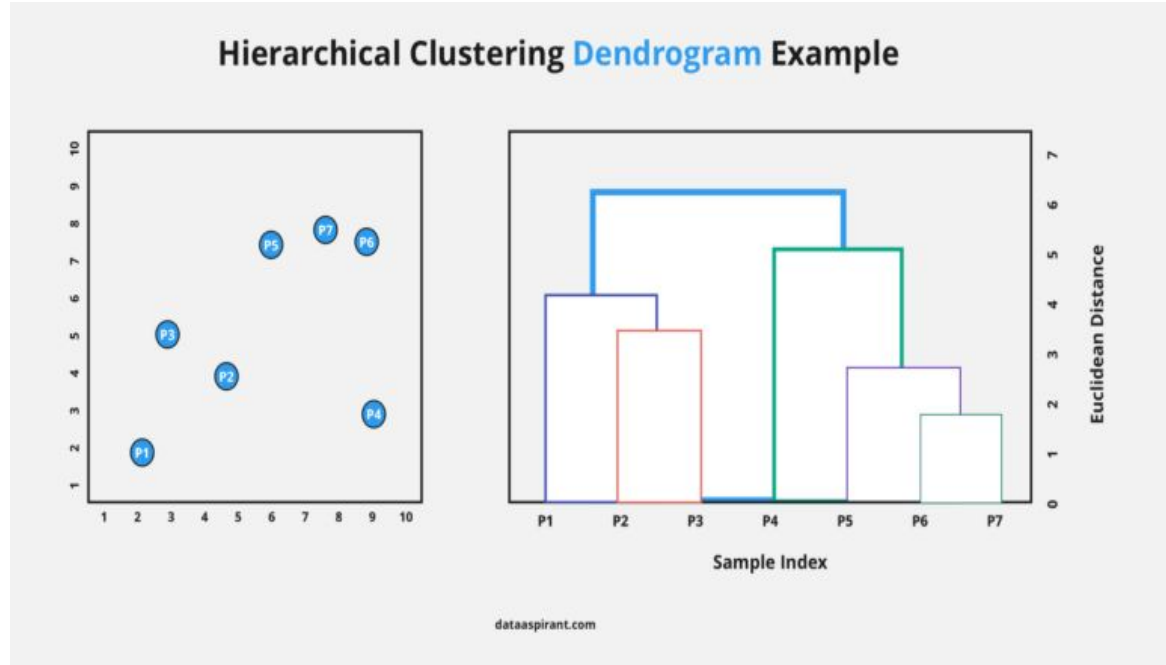


Step 4

Repeat 'Step 3' until you are left with **only one cluster**.

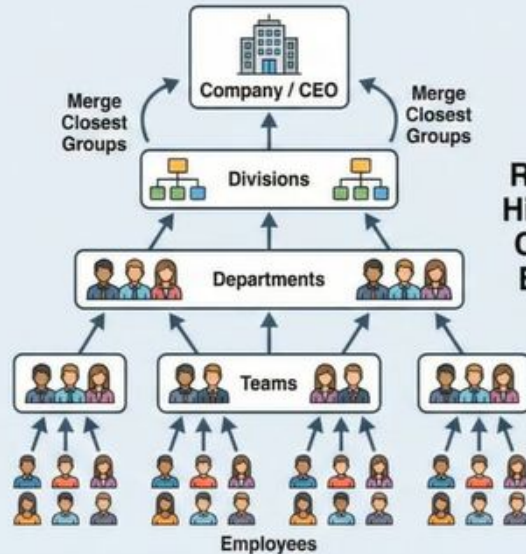


Once all the clusters are combined into a big cluster. We develop the Dendrogram to divide the clusters.

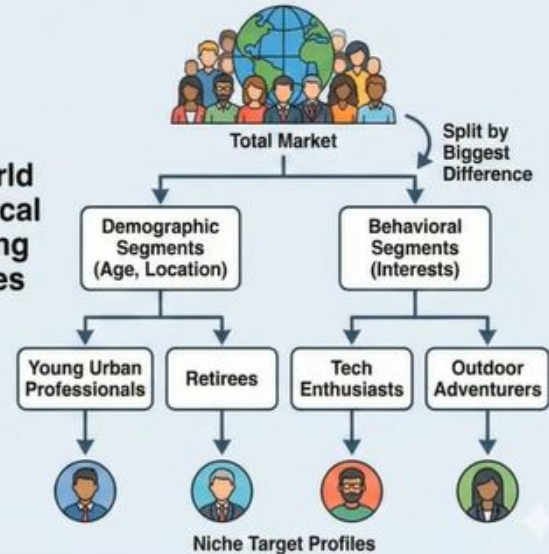


Real-World Hierarchical Clustering Examples

Agglomerative: Corporate Hierarchy (Bottom-Up)



Divisive: Market Segmentation (Top-Down)



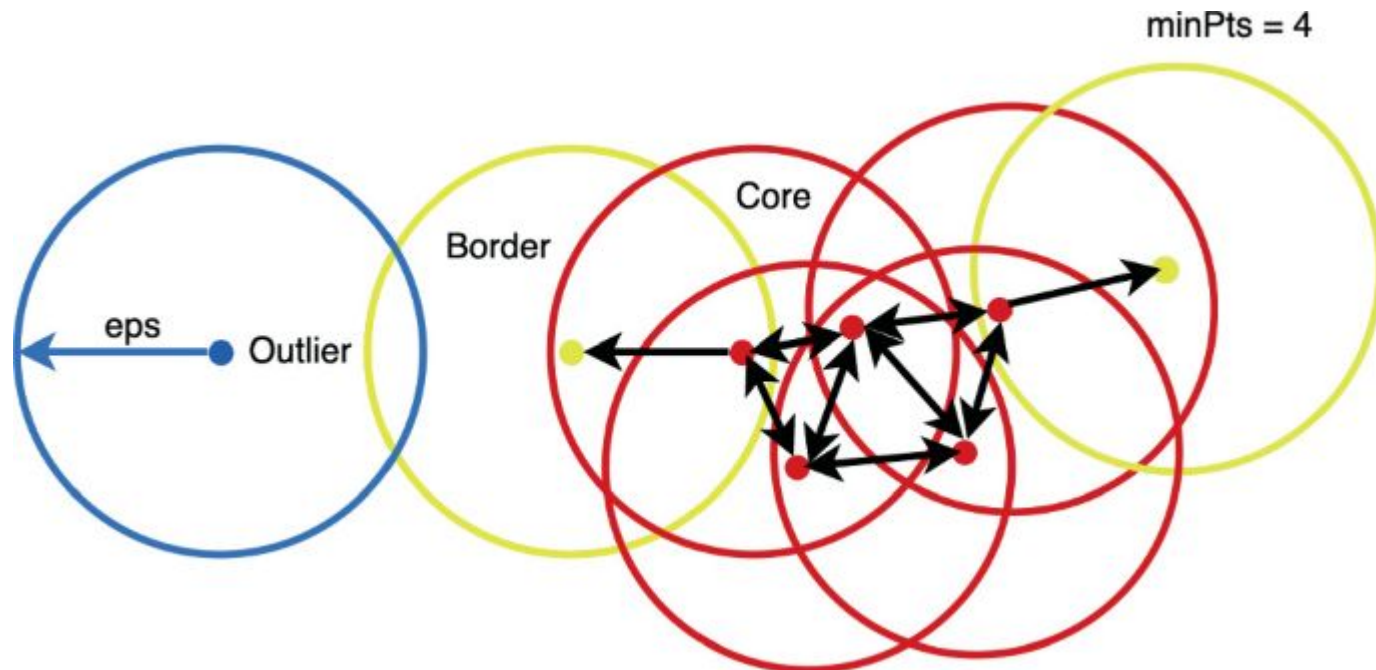
Real-World
Hierarchical
Clustering
Examples

DBSCAN

In the **DBSCAN** (Density-Based Spatial Clustering of Applications with Noise) algorithm, data points are classified into specific categories based on how tightly packed they are with their neighbors.

To determine the classification of a point, DBSCAN relies on two key hyperparameters:

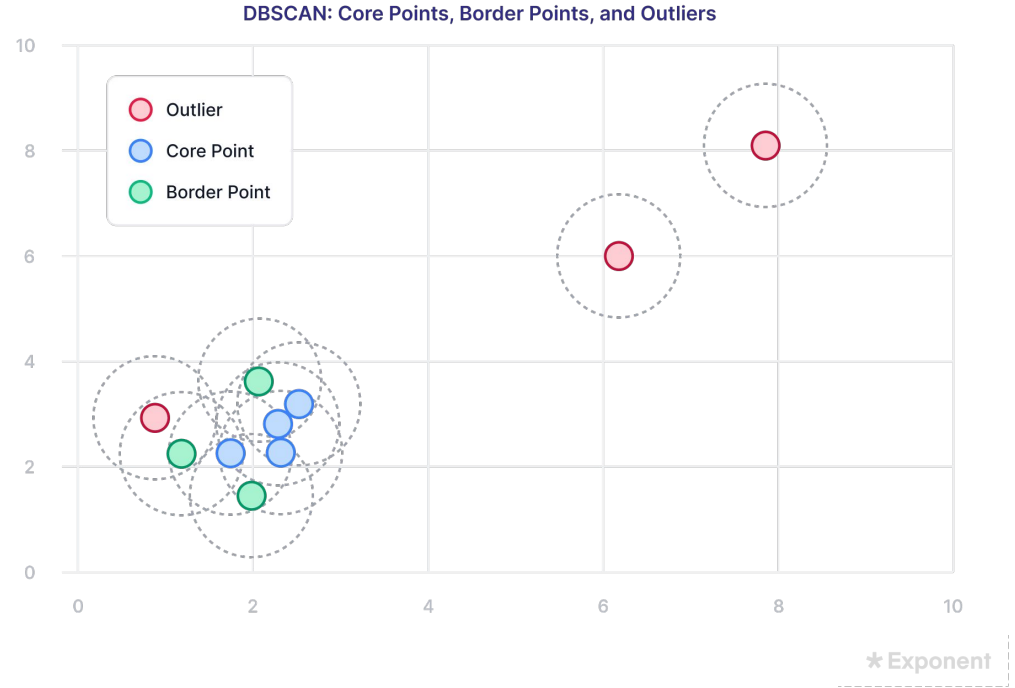
- **ϵ (Epsilon):** The maximum radius of the neighborhood around a point.
- **MinPts:** The minimum number of points required within the ϵ -neighborhood to form a dense region.



1. Core Point

2. Non-Core Points

- Border / Adjacent Point
- Outlier / Noise Point



DBSCAN Explained

Step-by-Step with a Numerical Example

DBSCAN Parameters



Epsilon (ϵ)

The maximum distance between two points for them to be considered part of the same cluster (radius of the circle).



Minimum Points (MinPts)

The minimum number of points within ϵ distance to form a dense region (including the point itself).



In this example: $\epsilon = 1.9$, MinPts = 4

Dataset: 12 Points (x, y)

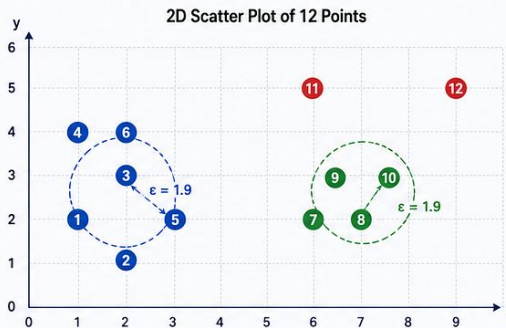
Point	x	y	Point	x	y
P1	1	2	P7	6	2
P2	2	1	P8	7	2
P3	2	3	P9	6	3
P4	1	4	P10	7	3
P5	3	2	P11	5	5
P6	2	4	P12	8	5

Final Clusters

- Cluster 1 (Blue): P1, P2, P3, P11
- Cluster 2 (Green): P4, P5, P6, P7, P8
- Cluster 3 (Orange): P10
- Noise (Red): P9, P12



DBSCAN finds clusters of arbitrary shape and handles noise effectively!



Point Types in DBSCAN

- Core Point**
At least MinPts points within ϵ distance (including itself).
- Border Point**
Within ϵ distance of a core point but has fewer than MinPts points in its neighborhood.
- Noise Point**
Neither a core point nor a border point.

Cluster 1 (Blue) Cluster 2 (Green) Cluster 3 (Orange) Noise (Red)

Step-by-Step Process

- Set Parameters:**
 $\epsilon = 1.9$ (points within 1.9 units are neighbors), MinPts = 4
- Calculate Distances:**
Use Euclidean distance:
$$d(p, q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: Distance between P1 (1,2) and P2 (2,1):
$$d = \sqrt{(2-1)^2 + (1-2)^2} = \sqrt{2} \approx 1.41 \leq 1.9 \rightarrow \text{Neighbors}$$
- Identify Core Points:**
Count neighbors within ϵ (including the point itself). If count $\geq$ MinPts $\rightarrow$ Core Point.
- Form Clusters:**
Start from a core point and include all density-reachable points.
- Identify Border & Noise Points:**
Border: within ϵ of a core point but not a core point.
Noise: not within ϵ of any core point.

Neighbors within $\epsilon = 1.9$ (including the point itself)

Point	Neighbors (IDs)	Count	Classification
P1	1, 2, 3, 4	4	Border Point
P2	1, 2, 3, 11	4	Core Point
P3	1, 2, 3, 4, 5, 6	6	Core Point
P4	1, 3, 4, 6	4	Core Point
P5	2, 3, 5	3	Border Point
P6	3, 4, 6	3	Border Point
P7	4, 5, 6, 7, 8	5	Core Point
P8	7, 8, 9, 10	4	Core Point
P9	8, 9, 10	3	Border Point
P10	8, 9, 10	3	Border Point
P11	2, 11, 12	3	Border Point
P12	11, 12	2	Noise Point



Set ϵ & MinPts



Find Neighbors



Classify Points



Form Clusters



Identify Noise



